

PHARMATHON PROBLEM STATEMENTS

1. Personalized Medicine & AI in Drug Development Problem: Traditional drug development is time-consuming and costly, often failing to address individual patient variability. How can AI and machine learning be leveraged to create a personalized drug formulation platform that predicts optimal drug dosages based on patient genomics and medical history?

2. Smart Drug Delivery Systems for Chronic Diseases Problem: Many patients struggle with adherence to long-term medication regimens due to frequent dosing and side effects. Can a smart drug delivery system be designed using nanotechnology, 3D printing, or biosensors to ensure controlled and sustained drug release for conditions like diabetes, hypertension, or cancer?

3. AI-Driven Drug Repurposing for Rare Diseases and Drug Resistance Problem: Drug development for rare diseases is slow due to limited patient populations and high costs. How can AI and big data analytics be used to identify existing drugs that could be repurposed for rare and neglected diseases?

4. Blockchain for Pharmaceutical Supply Chain Transparency Problem: Counterfeit drugs are a significant global health threat. Can blockchain technology be utilized to track and verify the authenticity of pharmaceuticals from manufacturing to patient consumption?

5. AR/VR for Pharmacy Education & Patient Counseling Problem: Pharmacy students and healthcare professionals often lack access to hands-on training in complex drug interactions and patient counseling. Can augmented reality (AR) or virtual reality (VR) be used to create interactive simulations for pharmaceutical education and patient engagement?

6. AI-Powered Digital Assistant for Medication Adherence Problem: Poor medication adherence leads to increased hospitalizations and treatment failures. Can an AI-driven digital assistant (chatbot/app) be developed to remind, educate, and monitor patients for improving medication compliance?

7. 3D-Printed Personalized Implants for Bone Tissue Engineering Problem:

Traditional bone grafts have limitations in customization and compatibility. Can 3D printing be used to develop personalized bone implants with drug-loaded scaffolds for better tissue regeneration?

8. Sustainable Pharmaceutical Packaging & Waste Management Problem:

Pharmaceutical waste, including expired drugs and packaging materials, poses environmental hazards. Can an eco-friendly pharmaceutical packaging or drug disposal system be developed to reduce medical waste?

9. AI-Driven Early Detection of Ocular Diseases Problem: Many eye diseases, such as glaucoma and keratitis, progress unnoticed until they cause significant damage. Can an AI-powered diagnostic tool be created to detect early-stage ocular diseases using smartphone-based retinal imaging or biosensors?

10. Smart Wound Healing Patches with Drug Delivery Problem: Chronic wounds, especially in diabetic patients, are difficult to treat and prone to infection. Can a smart wound dressing be developed that monitors healing and delivers antimicrobial or regenerative drugs in a controlled manner?

11. Error-Prone Manual Record-Keeping in Labs Problem: Manual recording of experimental results leads to errors, data loss, and lack of reproducibility in lab reports. Can a smart digital lab notebook (AI/Blockchain-based) be developed to automate result recording, analysis, and data validation?

12. Difficulty in Understanding Complex Analytical Techniques Problem:

Techniques like spectroscopy, chromatography, and dissolution testing require deep theoretical and practical understanding, which many students find challenging. Can AI-based interactive modules, AR/VR labs, or real-time simulation tools be created to enhance understanding?

13. Poor Handling and Disposal of Chemical & Biological Waste Problem:

Pharmacy labs generate a significant amount of chemical and biological waste, but students often lack proper training in safe handling and disposal. Can a smart waste management system be developed to classify, track, and ensure eco-friendly disposal of lab waste?

14. Limited Practical Exposure to Personalized Medicine & 3D Printing

Problem: With the rise of pharmacogenomics and 3D-printed personalized medicines, pharmacy students need more exposure to these emerging fields. Can an AI-powered tool or simulation model be developed to train students in personalized drug formulation and bioprinting applications?

15. Lack of Real-Time Guidance During Practical Sessions Problem: Students often perform experiments without instant expert supervision, leading to mistakes and incomplete learning. Can an AI-driven voice assistant or AR-based lab guide be developed to provide real-time step-by-step guidance and troubleshooting?

16. Inefficiency in Drug Stability & Quality Control Experiments Problem: Stability studies require longer time durations, and students often don't get to see real-time degradation results. Can a predictive AI model or simulation-based tool be designed to simulate real-time stability testing outcomes?

17. Insufficient Training in Regulatory Compliance & Good Laboratory Practices (GLP) Problem: Pharmacy students often lack practical exposure to regulatory guidelines (FDA, ICH, WHO) in lab settings. Can an interactive AI-based compliance checker be developed to train students on regulatory documentation, error detection, and GLP adherence?